steps in the next section.

5. Spectral Line Mapping

5.1 OTF Mapping Tutorial Overview

5.1.1 Introduction

For most maps at the LMT we use the technique of "on-the-fly" (OTF) mapping. In this procedure, the antenna is scanned over the source and data are collected at a high rate. The scanning pattern results in an irregular sampling of the positions on the sky, and so to make a final, science-grade map, we must use the OTF data to estimate the value of the map at specific, regularly spaced, grid points that also account for spatially redundant information.

One might imagine that using a "gridding" procedure to make the map would be an obvious step. However, gridding procedures, such as those in python, are actually interpolation algorithms which seek to interpolate between the data points closest to the grid points to make the estimate. This works well in cases of high signal-to-noise ratio, but in the typical case of low signal-to-noise astronomy data, we'd like to make use of more data samples to find an estimate of the map at the specified grid points.

A method for estimating values on a regular grid using OTF mapping has been developed and is in common use in the OTF reduction applications in use at all single dish telescopes and at ALMA for single dish mapping. The technique is described in a paper by Mangum, Emerson, and Griesen (A&A 474 679-687 2007) as well as in other papers describing OTF systems (see e.g. Sawada et al. PASJ 60 445-455 2008 and FCRAO internal memos by M. Brewer). The basic principles of the method are straightforward, but different groups describe the problem in different ways which can make matters confusing. Thus, I offer this short tutorial. A set of simulations of the procedures are presented in Appendix D.

5.1.2 OTF Mapping Basics

Let's call the true brightness distribution of a source on the sky *I(x,y)*, where *x* and *y* are the sky coordinates. When we make a map with the LMT, we do not measure *I*, but the convolution of *I* with the beam pattern of the telescope, which I will call *B*. Lets call the result I_{obs} and then we may write the convolution:

$$I_{obs}(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} B(x-x',y-y')I(x,y)dx'dy' \qquad \text{<1>}$$

I'll write this another way using the convolution operation "*" to make life easier:

$$I_{obs} = B*I \qquad \text{<2>}$$

From the convolution theorem I know that if I have two functions *f* and *g* then

$$FT(f*g) = FT(f) \cdot FT(g) \qquad \text{<3>}$$

where *FT* denotes a Fourier Transform and we represent multiplication with the symbol "·" to distinguish it from the convolution operation. Therefore, for our mapping problem, we can write:

$$FT(I_{obs}) = FT(B) \cdot FT(I) \qquad \text{<4>}$$

The Fourier Transform of the map we can make, I_{obs}, is the product of the Fourier Transform of the true map and the Fourier Transform of our beam pattern *B*. Now the beam power pattern itself is related to the Fourier Transform of the electric field in the antenna aperture, and to make a long story short, FT(*B)* has a maximum spatial frequency of D/λ. Since FT(*B)* multiplies the spatial frequency information of our true map, *I*, it follows that the observed map, I_{obs}, can have no spatial frequency information at frequencies higher than D/λ.

5.1.3 Receiver Noise in the Map

Noise from the receiver will cause the signal we observe to fluctuate rapidly so that if we scan across the source we will see changes in signal that are due to receiver noise and not due to real spatial brightness variations in the source itself. Therefore, if I take the Fourier Transform of the observed map to see its spatial frequency information, I would typically find power apparent at spatial frequencies greater than D/λ. We know that the source brightness can't really change on very fine spatial scales (high spatial frequencies) in our map; these fluctuations are not an actual property of the source. Therefore, we'd like to use that information to eliminate contributions of noise at impossible spatial frequencies from the map.

An easy way to remove the effect of noise at impossible spatial frequencies is to: (1) take the Fourier Transform of our data map, FT(I_{obs}); (2) multiply FT(I_{obs}) by a function (we'll call it Π) with value 1 for all spatial frequencies less than D/λ and 0 for all spatial frequencies greater than D/λ; and then (3) Fourier Transform that result back to get a new estimate of I_{obs} without any noise which we know is not a real property of the source map. Using the convolution theorem, we can see:

$$FT(FT(I\,obs) \cdot \Pi) = I_{obs} * FT(\Pi) \qquad \text{<5>}$$

and the operation we described is just a convolution of the Fourier Transform of the Π function with our observed map I_{obs}. The Π function just has unit value out to some radius (D/ λ) and its Fourier Transform is well known:

$$FT(\Pi) = \pi\left(\frac{D}{\lambda}\right)^2 \cdot 2\,\frac{J_1(2\pi r\, D/\lambda)}{2\pi r\, D/\lambda} \qquad \text{<6>}$$

where r is the offset in sky position. If I remove the normalization term and consider r to be scaled so that r = r' λ/D, then our convolution function becomes

$$C(r') = 2\frac{J_1(2\pi r')}{2\pi r'} = 2\, jinc(2\pi r') \qquad \text{<7>}$$

5.1.4 Resampling the Map

If I convolve my observed map, I_{obs}, with the function *C* defined above, then I know that I have not removed any features from the map that could be physically present. Now I just need to be sure to sample the map, according to the Nyquist sampling theorem, at a rate that corresponds to twice the maximum spatial frequency that is possible in the data. This corresponds to a spatial frequency of 2D/λ, or a spacing of samples on the sky of λ/2D. I can sample on a finer grid than this as well, but it will not change the spatial frequency content (or resolution) of the map.

So, the above arguments may be used to explain OTF mapping procedure. We convolve the actual, irregularly spaced, data with function C and then sample the result on a new, regular, grid with grid spacing of λ/2D in order to preserve all the spatial frequency information that it is possible to have. In so doing, we are usually combining several data points to make an estimate of I_{obs} at each grid point. This can lead to an improved estimate of the map value of that grid point over what would have been obtained from a single sample.

5.1.5 Nearest Neighbor Gridding

The *jinc* filter approach sounds complicated, and people often wonder whether a simpler approach will work. A common alternative approach to making maps from a set of data points is to divide the sky into "cells" and average all data points that fall within a cell. What happens when you do this?

Following the above reasoning, it should be apparent that the nearest neighbor gridding procedure is actually a convolution of your map data with a box function with unit amplitude within the map cell. In spatial frequency this means multiplying the spatial frequencies of the data map by the Fourier Transform of the box. Let c be the width of the cell in units of λ/D, and u and v be the spatial frequencies in units of D/λ, then the Fourier Transform of the cell box, $FT_{cell}(u,v)$, may be be written:

$$FT_{cell}(u,v) = \frac{\sin(\pi c u)}{\pi c u} \cdot \frac{\sin(\pi c v)}{\pi c v} \qquad \text{<8>}$$

Thus, this procedure multiplies the above Fourier Transform of the cell box by the Fourier Transform of the map of the data. Since the this function has values at spatial frequencies higher than D/λ, the resulting map will include noise at spatial frequencies than can't exist. Therefore, the OTF filtering approach has been developed, and in this approach, we define a filter that eliminates the high spatial frequencies effectively. We will do more comparisons of the "nearest neighbor" and OTF filter approaches after dealing with some practical issues in optimizing the OTF filter.

5.1.6 Dealing with Practical Problems

The main problem with the simplified OTF filter algorithm outlined above is that the *jinc* function (like the *sinc* function of the nearest neighbor algorithm) has lobes with significant values to great distances from the convolution point. Figure 1 shows the *jinc* function sampled out to 12 times λ/D. The spatial frequency response of the *jinc* convolution with three cutoff values (RMAX) of 3 λ/D, 6 λ/D, and 12 λ/D is shown in the right-hand panel. It is clear that the sidelobes of the *jinc* function lead to a nonuniform spatial frequency response.

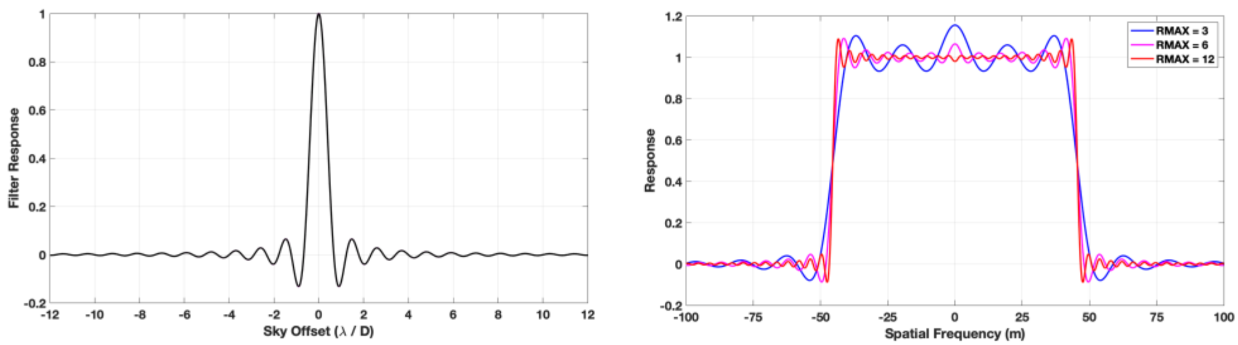

Figure 1 – (left) plot of the jinc function assuming uniform weight of spatial frequencies out to the diameter of the antenna. (right) Spatial frequency response for jinc truncated at different maximum distances (in units of λ/D).

Since It is impractical to carry out the convolution to infinite distance, practical filters have been developed to reduce the *jinc* sidelobes to a low level by multiplying them with a gaussian and/or some other function so that the sidelobe level is close to zero by the time the maximum distance for the convolution is reached, usually a few times λ/D. The equation (9) below gives a common function used in the FCRAO OTF mapping software. Figure 2 illustrates what the additional terms do as the original *jinc* is successively multiplied by a gaussian and then a second *jinc* chosen to reach its first zero at precisely the convolution cutoff radius, RMAX.

$$c(r') = 2\ jinc\left(\frac{2\pi r'}{a}\right)\ \exp\left(-\left(\frac{2r'}{b}\right)^{c}\right)\ 2\ jinc\left(\frac{3.831706\, r'}{RMAX}\right) \qquad \text{<9>}$$

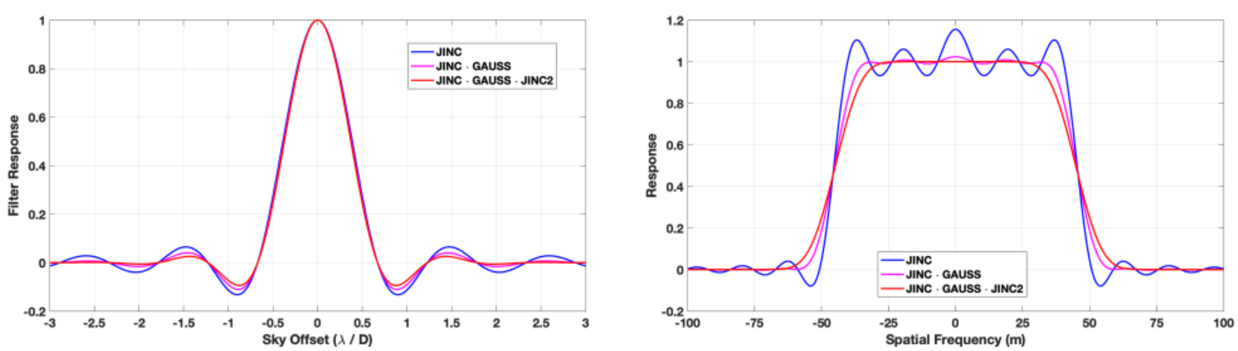

Figure 2 – (left) plot of jinc function (blue); product of jinc and gaussian function (magenta); and product of jinc, gaussian, and a second jinc scaled so that it's first zero is at the cutoff of the convolution function. In this case, the cutoff is at RMAX=3. (right) Spatial frequency response of the three filters. For these plots, the otf "a" parameter is 1.1, the otf "b" parameter is 4.75, and the otf "c" parameter is 2.

Each successive multiplication of the gaussian and the second *jinc* function has the benefit of reducing the first *jinc's* sidelobes and thereby reducing ripples in the spatial frequency response of the filter. However, each new function convolves the spatial frequency response with its Fourier Transform. These additional convolutions lower the response to the highest spatial frequencies and increase the response to noise that lies just outside the maximum spatial frequency that is possible in the map. Usually, a lower response at the highest spatial frequency is justified since the antenna illumination is tapered and the contribution from the edges of the dish is downweighted anyway. The response from just outside the maximum spatial frequency is tolerated since the process does succeed in removing responses at even higher spatial frequencies that would exist without the additional application of the gaussian and second jinc function.

5.2 Data Processing Steps

5.2.1 Overview

An overview of the process to create an OTF map is given in Figure __. The initial steps follow the procedure to reduce the raw IFProc and Roach files to create individual spectra for each "dump" of the spectrometer. Then the spectra may require baseline removal or line fitting to establish the properties to be mapped in addition to the values of the spectral channels.

In the LMT system, all spectra in an individual observation are taken with the same configuration of the spectrometer, so we may store all data in a two-dimensional data structure with one dimension representing the sequence of the observation and the second dimension representing the spectrum itself. Each spectrum in the sequence has timing and other information that will be required to make the map. All spectra correspond to a single set of velocities or frequencies depending on the processing at the "line" stage. The set of all spectra corresponding to a single observation can be written to a "SpecFile" for later analysis.

Next in the procedure is to read all reduced spectra from the SpecFIle and use the OTF filter to accumulate them into a regular, three-dimensional grid. The result of this procedure is to write the final data cube of information into a new file for viewing and analysis.

Finally, given multiple maps of the same region, it is useful to be able to combine the data cube from each map into a new data cube to improve the signal-to-noise ratio of the map.